

PORTFOLIO CASE STUDY

Detecting Registry Run Key Persistence with Sysmon & Splunk

Project: Enterprise Endpoint Threat Detection Lab **Simulation:** Persistence Attack — Registry Run Keys (T1547.001) **Analyst:** Olayinka Oyetade **Tools:** Sysmon · Splunk Enterprise · Windows Event Logs · SPL **Outcome:** Full detection pipeline validated — attack to alert in < 5 minutes

THE PROBLEM

Attackers who successfully compromise a Windows machine face one core challenge: **how do they stay on the machine after a reboot?**

One of the most common answers is Registry Run keys — registry locations that Windows reads automatically at every login to launch programs. By writing a malicious entry there, an attacker ensures their malware restarts every time the victim logs in — even if the file is temporarily removed by antivirus.

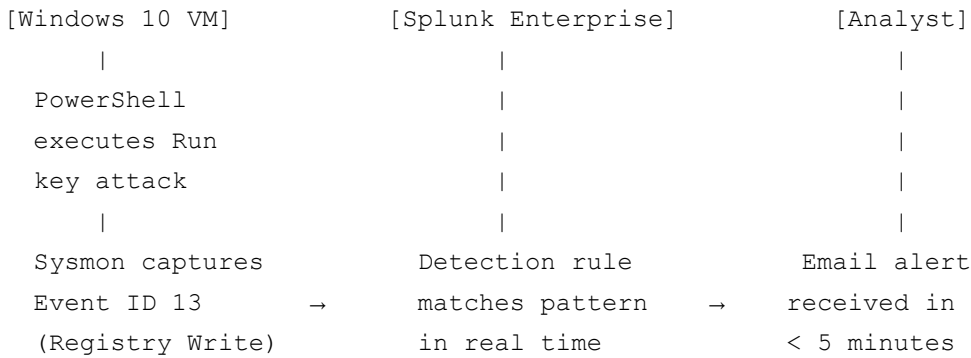
This technique is used by some of the most prolific malware families in the world:

- **Emotet** — banking trojan responsible for billions in damages
- **TrickBot** — credential stealer and ransomware loader
- **Remcos RAT** — remote access trojan sold as "legitimate" software
- Countless ransomware families that need to survive reboots before encrypting files

Most organizations detect this too late — or not at all. This project demonstrates how a properly configured Sysmon + Splunk pipeline catches it in real time.

WHAT I BUILT

A fully functional SOC home lab replicating enterprise endpoint detection:



Stack:

- Windows 10 VM (VMware) — the endpoint being monitored
- Sysmon v14 — kernel-level telemetry collection
- Splunk Enterprise — SIEM for log ingestion and detection
- Splunk Add-on for Microsoft Sysmon — field extraction
- SPL (Search Processing Language) — detection query and alerting

THE ATTACK

Simulated attacker gains code execution and runs:

```
New-ItemProperty -Path "HKCU:\Software\Microsoft\Windows\CurrentVersion\Run" `
    -Name "WindowsUpdate" `
    -Value "C:\Users\Public\malware.exe" `
    -PropertyType String `
    -Force
```

This creates a registry entry disguised as a legitimate Windows process ("WindowsUpdate") that executes `malware.exe` every time the user logs in. No admin rights required.

THE DETECTION

Sysmon captures the registry modification as **Event ID 13 (Registry Value Set)** with full forensic context:

Field	Value

Event ID	13
Source Process	powershell.exe
Registry Key	HKCU...\CurrentVersion\Run\WindowsUpdate
Value Written	C:\Users\Public\malware.exe
User	WINDOWSVM10\mrdaniel198

SPL Detection Query:

```
index=endpoint EventCode=13 "powershell.exe" "CurrentVersion\\Run"
| table _time, host, User, Image, TargetObject, Details
| sort -_time
```

This query detects the attack technique — not a specific malware signature. That means it catches **any** malware using this persistence method, regardless of the payload name or content.

KEY RESULTS

Metric	Result
Time to detect	< 5 minutes from execution to alert
False positive rate	Low — PowerShell + Run key filter is highly specific
Detection method	Behavioral (technique-based), not signature-based
Alert type	Automated email notification
MITRE coverage	T1547.001 fully covered
Pipeline validated	Attack → Sysmon → Splunk → Alert → Analyst 

WHY THIS MATTERS FOR SOC WORK

This is what SOC analysts do every day.

The workflow replicated here — simulate attack, analyze telemetry, refine detection logic,

validate alerting — is the same workflow used by detection engineers at enterprise security teams.

The specific skills demonstrated:

SIEM Proficiency: Writing and tuning SPL queries to detect real attack patterns, not just running pre-built searches.

Detection Engineering: Understanding the difference between signature detection (brittle, easily bypassed) and behavioral detection (technique-based, durable across malware variants).

Alert Tuning: Iterating on detection rules to reduce false positives — a critical skill that separates effective SOC analysts from those who create alert fatigue.

MITRE ATT&CK Framework: Mapping detections to the industry-standard attack framework used by SOC teams, threat intelligence teams, and red teams across the industry.

Incident Documentation: Producing professional incident reports, detection documentation, and lessons learned — not just running queries.

HOW I EXPLAIN THIS IN AN INTERVIEW

"I built a Windows threat detection lab using Sysmon and Splunk, then simulated a real attacker persistence technique — specifically Registry Run Key modification, which is MITRE ATT&CK T1547.001. This technique is used by Emotet, TrickBot, and dozens of ransomware families.

I wrote a Splunk detection rule using SPL that catches the behavior — not the specific malware file — so it works against any variant of this technique. The trickier part was tuning the rule. My first version generated a lot of false positives from legitimate Windows processes also using Run keys. I refined it by filtering on the source process, which dramatically improved precision.

The full pipeline — attack execution to automated email alert — runs in under 5 minutes. That's the kind of MTTD performance SOC teams care about."

EVIDENCE ARTIFACTS

Artifact	Description

screenshots/sysmon-event13-detection.png	Splunk result showing Event ID 13 with full forensic fields
screenshots/splunk-alert-fired.png	Saved search alert triggered in Splunk
screenshots/email-notification.png	Automated email alert received in analyst inbox
queries/detection-rule-001.spl	Production-ready SPL detection query
IR-2026-001-incident-report.md	Full incident report
DET-001-detection-rule-doc.md	Detection engineering documentation
LL-2026-001-lessons-learned.md	Post-simulation lessons learned

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